

From Analytic Spectral Collapse to Sobolev Spectral Disks

Exact regularity thresholds and sharp endpoint laws
for Rvachev–Thue–Morse transfer operators

Research manuscript prepared for Vladimir Reshetnikov
OpenAI ChatGPT

29 September 2026 (Pacific time)

Abstract

The ProveIt manuscript on spectral collapse and alternating Rvachev RMS asymptotics asks how its analytic spectrum changes at finite smoothness. We answer the periodic Sobolev part of that question, including logarithmic refinements, and the nonperiodic interval problem at every nonnegative integer Sobolev order. For a trigonometric polynomial mask and an integer dilation, a positive eigenfunction of the squared-mask transfer operator gives the exact Fredholm essential spectrum: a closed disk. The complete spectrum is that disk together with the spectrum of an explicit finite Fourier matrix. We prove the required positivity hypothesis for every even sine power.

For the dyadic quadratic sine mask the disk radius is $2^{-s}\sqrt{1+\sqrt{17}}/4$. The eigenvalues $1/2$ and $-1/4$ become isolated exactly above two regularity thresholds separated by one derivative. An exact Fourier-cutoff identity links the leading Thue–Morse correlation functional to the subleading Stern-polynomial functional $B_k(-2)$. Closed quadratic-energy formulas determine their precise Sobolev and logarithmic continuity thresholds. They also give two-sided, sharp operator norms for the scalar moment remainder. At the subleading critical index, a logarithmic weight of exponent $\beta > 1/2$ recovers a uniform two-term scalar asymptotic even though the subleading eigenvalue remains on the essential spectral boundary. No corresponding operator-norm little-oh remainder is possible.

All infinite-dimensional conclusions are proved analytically. An accompanying standard-library Python program checks finite identities with exact rational and quadratic-field arithmetic. This is an unrefereed research extension, not a new Lean formalization or a claim of established global priority.

Keywords. Transfer operator; essential spectrum; Sobolev regularity; Thue–Morse correlations; Stern polynomials; Riesz product; critical asymptotics.

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1 The repository question and the contribution

1.1 A specific, delimited research target

The Fabius-function branch of ProveIt contains a Fourier-decay synthesis and a later manuscript entitled *Spectral Collapse and Alternating RMS Asymptotics for the Rvachev Up-Function* [1, 2]. The later manuscript proves, on analytic Hardy disk spaces, that the quadratic dyadic transfer operator has spectrum $\{0, 1/2, -1/4\}$. Its research question *Spectra at finite smoothness* asks for the spectrum or essential spectral radius on finite-smoothness spaces, the isolation of the analytic eigenvalues, and the effect of unmatched endpoint jets. It explicitly warns against transferring an analytic determinant to such spaces.

This paper answers that question on $H^{s,\beta}(\mathbb{T})$ for all real s, β , and on $H^r(0, 1)$ for every integer $r \geq 0$. The answer differs fundamentally from analytic spectral collapse: there is an entire essential spectral disk, not merely a finite set. The remaining Hölder, C^r , and fractional nonperiodic interval questions are not claimed to be resolved.

The comparison is fixed by the repository commit

b2aed2d23125006c48d63fc97aef0d64b67becb1.

The exact source path, inspected material, and source identifiers are recorded in section A. No repository file was modified in preparing this article. In particular, a result described here as proved is a claim about the mathematical argument in this article, not a claim that the repository’s Lean development already contains it.

1.2 What is already known, and what is supplied here

Transfer operators with sine and cosine weights have an established literature. Chen and Volkmer study precisely this family on the circle, including explicit dyadic computations [3]. Disk spectra for weighted isometries and their connection with transfer operators are part of a broader theory studied by Bardadyn and Kwaśniewski [4]. Thue–Morse correlations, their Riesz-product origin, and questions about their means are also established objects [5]. The recurrence defining Stern polynomials is classical; we use the normalization of Ulas and Ulas [6].

The present work does not claim to discover those operators, sequences, or the general possibility of a disk spectrum. Its contribution is the self-contained combination of an exact Sobolev essential disk with a complete finite-matrix classification of the outliers, followed by a sharp critical-regularity analysis of the two distinguished dyadic coefficient functionals. The exact Fourier-tail formula yields a scalar asymptotic at an embedded spectral boundary, together with an optimal logarithmic threshold. These statements provide a concrete answer to the inspected repository’s finite-smoothness question.

The literature checks establish relevant antecedents but are not an exhaustive priority investigation. Some retrieved sources were available only through primary-source abstracts and metadata. Accordingly, the results are presented as a proved research extension relative to the specified repository target, not as a certified first occurrence of every specialization in the literature.

1.3 The principal dyadic conclusion

Let

$$(\mathcal{L}f)(x) = \frac{1}{2} \left[\sin^2\left(\frac{\pi x}{2}\right) f(x/2) + \cos^2\left(\frac{\pi x}{2}\right) f((x+1)/2) \right]. \quad (1.1)$$

Put

$$\gamma = \frac{\sqrt{1 + \sqrt{17}}}{4}, \quad s_0 = \frac{1}{2} \log_2 \left(\frac{1 + \sqrt{17}}{4} \right), \quad s_1 = s_0 + 1. \quad (1.2)$$

The exact classification proved below is

$$\sigma_{\text{ess}}(\mathcal{L}; H^{s,\beta}) = \overline{D}(0, \gamma 2^{-s}), \quad \sigma(\mathcal{L}; H^{s,\beta}) = \overline{D}(0, \gamma 2^{-s}) \cup \{1/2, -1/4\}. \quad (1.3)$$

Here essential spectrum means *failure of Fredholmness*. The real logarithmic exponent β does not change this spectral set. It does, however, decide whether the leading and subleading coefficient functionals are continuous at $s = s_0$ and $s = s_1$.

For the sine-product moments

$$W_n(x) = \prod_{j=0}^{n-1} \sin^2(\pi 2^j x), \quad I_n(f) = \int_0^1 f(x) W_n(x) dx, \quad (1.4)$$

we construct functionals μ, D_+, D_- such that

$$I_n(f) = 2^{-n} \mu(f) + \frac{(-1/4)^n}{3} (D_+ + D_-)(f) + R_n(f). \quad (1.5)$$

At the critical subleading regularity,

$$\|R_n\|_{(H^{s_1,\beta})^*} \asymp_\beta 4^{-n} (1+n)^{1/2-\beta} \quad (\beta > 1/2). \quad (1.6)$$

Thus a uniform scalar remainder $o(4^{-n})$ survives at the essential spectral boundary, while an operator-norm remainder $o(4^{-n})$ cannot. This distinction is one of the main reasons to study both the spectrum and the actual moment functional rather than substitute one question for the other.

2 Spaces, masks, and the classification theorem

2.1 Fourier conventions and logarithmic Sobolev spaces

Write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, $e_k(x) = \exp(2\pi i k x)$, and $\widehat{f}(k) = \int_0^1 f(x) e^{-2\pi i k x} dx$ for integrable f . For real s, β , define $H^{s,\beta}(\mathbb{T})$ as the Hilbert completion of trigonometric polynomials in the norm

$$\|f\|_{s,\beta}^2 = \sum_{k \in \mathbb{Z}} \langle k \rangle^{2s} [\log(e + |k|)]^{2\beta} |\widehat{f}(k)|^2, \quad \langle k \rangle = (1 + k^2)^{1/2}. \quad (2.1)$$

The notation H^s means $H^{s,0}$. For negative s , the completion is naturally a space of periodic distributions. Every formula initially stated on trigonometric polynomials extends by boundedness when the relevant norm estimate is established. Pairings with distributions are complex-linear in f ; they need not be Hilbert inner products.

Let $b \geq 2$ be an integer and let

$$w(x) = \sum_{\ell=-D}^D w_\ell e_\ell(x) \neq 0, \quad (\mathcal{L}_{b,w} f)(x) = \frac{1}{b} \sum_{a=0}^{b-1} (wf) \left(\frac{x+a}{b} \right). \quad (2.2)$$

The Fourier rule is

$$\widehat{\mathcal{L}_{b,w} f}(k) = \sum_{\ell=-D}^D w_\ell \widehat{f}(bk - \ell), \quad \mathcal{L}_{b,w} e_j = \sum_k w_{bk-j} e_k. \quad (2.3)$$

In particular, these operators are bounded on every $H^{s,\beta}$: only finitely many frequency shifts occur, and the ratio of the corresponding weights is bounded.

2.2 The finite matrix and the positive square-mask hypothesis

Set

$$d = \left\lfloor \frac{D}{b-1} \right\rfloor, \quad F_d = \text{span}\{e_j : |j| \leq d\}, \quad A_{b,w} = (w_{bk-j})_{-d \leq k, j \leq d}. \quad (2.4)$$

A coefficient outside $[-D, D]$ is understood to be zero. This matrix is exactly the restriction of $\mathcal{L}_{b,w}$ to F_d .

The hypothesis used in the general result is

$$h \in C(\mathbb{T}), \quad \min_{\mathbb{T}} h > 0, \quad \rho > 0, \quad \mathcal{L}_{b,|w|^2} h = \rho h. \quad (\text{P})$$

It is important that h be strictly positive, not merely nonnegative. A zero of h would invalidate the bounded similarity used below. We prove (P) for all masks $\sin^{2m}(\pi x)$ in section 5. For a general complex trigonometric mask, it remains an explicit hypothesis.

Theorem 2.1 (Exact Sobolev disk and finite outliers). *Assume (P). For all real s, β , set $r_s = b^{-s} \sqrt{\rho}$. Then*

$$\sigma_{\text{ess}}(\mathcal{L}_{b,w}; H^{s,\beta}) = \overline{D}(0, r_s), \quad (2.5)$$

$$\sigma(\mathcal{L}_{b,w}; H^{s,\beta}) = \overline{D}(0, r_s) \cup \sigma(A_{b,w}). \quad (2.6)$$

Every generalized eigenvector at an eigenvalue λ with $|\lambda| > r_s$ lies in F_d . The algebraic multiplicities and Jordan blocks at such λ agree with those of $A_{b,w}$.

Moreover, for every $n \geq 1$ and every compact operator K on $H^{s,\beta}$,

$$\left\| \mathcal{L}_{b,w}^n - K \right\|_{H^{s,\beta} \rightarrow H^{s,\beta}} \geq r_s^n. \quad (2.7)$$

The compact operator may depend on n .

The theorem identifies a *set-valued spectrum*, not only its outer radius. It does not identify the point, continuous, and residual spectra inside the Sobolev disk. On L^2 , the proof supplies infinite-dimensional eigenspaces for every point in the open disk. Compact perturbation preserves the Fredholm essential spectrum, but not those individual eigenspaces; we therefore make no analogous interior point-spectrum assertion at every s .

3 The exact L^2 disk

3.1 A similarity to a coisometry

Let M_g denote multiplication by a bounded function g . All adjoints in this section are taken in $L^2(\mathbb{T})$ with Lebesgue measure.

Lemma 3.1 (Square-mask identity). *For bounded h ,*

$$\mathcal{L}_{b,w}^* g = \overline{w} (g \circ T_b), \quad \mathcal{L}_{b,w} M_h \mathcal{L}_{b,w}^* = M_{\mathcal{L}_{b,|w|^2} h}, \quad T_b x = bx \pmod{1}. \quad (3.1)$$

Proof. Changing variables on each of the b branches gives $\int (\mathcal{L}_{b,w} f) \overline{g} = \int w f \overline{g \circ T_b}$, which proves the adjoint formula. For the second identity, evaluate $\mathcal{L}_{b,w} (h \overline{w} g \circ T_b)$ at x . The factor $g(x)$ does not depend on the chosen inverse branch and can be taken outside the sum. What remains is $\mathcal{L}_{b,|w|^2} h(x)$. $\square$

Lemma 3.2 (Coisometry and power bound). *Under (P), the operator*

$$C = \rho^{-1/2} M_{h^{-1/2}} \mathcal{L}_{b,w} M_{h^{1/2}} \quad (3.2)$$

is a coisometry: $CC^ = I$. Consequently,*

$$\|\mathcal{L}_{b,w}^n\|_{2 \rightarrow 2} \leq \left(\frac{\max h}{\min h} \right)^{1/2} \rho^{n/2} \quad (n \geq 0). \quad (3.3)$$

Proof. The square-mask identity gives $CC^* = \rho^{-1} M_{h^{-1/2}} M_{\rho h} M_{h^{-1/2}} = I$. Thus C^* is an isometry and $\|C\| = 1$. The similarity $\mathcal{L}_{b,w} = \sqrt{\rho} M_{h^{1/2}} C M_{h^{-1/2}}$ yields (3.3). $\square$

3.2 Why the whole disk is essential

Lemma 3.3 (Infinite branch defect). *The kernel of $\mathcal{L}_{b,w}$ on $L^2(\mathbb{T})$ is infinite-dimensional.*

Proof. A nonzero trigonometric polynomial has only finitely many zeros on $\mathbb{T}$. Choose a small interval $J \subset (0, 1)$ on which two branch weights $w(y_0(x))$ and $w(y_1(x))$, with $y_a(x) = (x + a)/b$, do not vanish. For arbitrary $g \in L^2(J)$, define f on these two branch intervals by

$$f(y_0(x)) = w(y_1(x))g(x), \quad f(y_1(x)) = -w(y_0(x))g(x),$$

and let f vanish on the other branches and outside these intervals. The two contributions to $\mathcal{L}_{b,w}f$ cancel. Because the weights are bounded away from zero after shrinking J , this construction embeds the infinite-dimensional space $L^2(J)$ into the kernel. $\square$

Proposition 3.4 (Complete essential disk on L^2). *Under (P),*

$$\sigma(\mathcal{L}_{b,w}; L^2) = \sigma_{\text{ess}}(\mathcal{L}_{b,w}; L^2) = \overline{D}(0, \sqrt{\rho}). \quad (3.4)$$

Every point of the open disk is an eigenvalue of infinite geometric multiplicity.

Proof. Let $E = \ker C$, which is infinite-dimensional by theorem 3.3 and the similarity. For $e \in E$, the vectors $(C^*)^j e$, $j \geq 0$, lie in mutually orthogonal copies of E . Indeed, if $j > k$, cancellation of k copies of $CC^* = I$ reduces the inner product to one involving a positive power of C applied to an element of $\ker C$, and it is zero.

For $|z| < 1$, the norm-convergent series

$$u_z(e) = \sum_{j=0}^{\infty} z^j (C^*)^j e \quad (3.5)$$

satisfies $Cu_z(e) = zu_z(e)$ and $\|u_z(e)\|^2 = \|e\|^2 / (1 - |z|^2)$. It is injective in e . Therefore $C - zI$ has infinite-dimensional kernel and is not Fredholm. The non-Fredholm set is closed, so it contains the closed unit disk. Since $\|C\| = 1$, there is no spectrum outside that disk. Similarity and scaling prove the claim for $\mathcal{L}_{b,w}$. $\square$

No abstract classification of all weighted shifts is needed for this argument. The branch defect and the positive square-mask eigenfunction supply the complete disk directly.

4 Compact conjugacy and the exclusion of hidden outliers

4.1 The compact remainder in a change of regularity

Let $U_{s,\beta} : H^{s,\beta} \rightarrow L^2$ be the unitary Fourier multiplier

$$U_{s,\beta} e_k = q_{s,\beta}(k) e_k, \quad q_{s,\beta}(k) = \langle k \rangle^s [\log(e + |k|)]^\beta. \quad (4.1)$$

Lemma 4.1 (Compact Sobolev conjugacy). *For every real s, β ,*

$$U_{s,\beta} \mathcal{L}_{b,w} U_{s,\beta}^{-1} = b^{-s} \mathcal{L}_{b,w} + K_{s,\beta}, \quad K_{s,\beta} \text{ compact on } L^2. \quad (4.2)$$

For every fixed positive integer n , it follows that

$$U_{s,\beta} \mathcal{L}_{b,w}^n U_{s,\beta}^{-1} = b^{-ns} \mathcal{L}_{b,w}^n + K_{s,\beta,n}, \quad K_{s,\beta,n} \text{ compact}. \quad (4.3)$$

Proof. In the Fourier rule, the entry that sends input frequency $bk - \ell$ to output frequency k is multiplied under conjugacy by $q_{s,\beta}(k)/q_{s,\beta}(bk - \ell)$. For each fixed ℓ ,

$$\frac{q_{s,\beta}(k)}{q_{s,\beta}(bk - \ell)} \longrightarrow b^{-s} \quad (|k| \rightarrow \infty). \quad (4.4)$$

The power part has this limit because $\langle k \rangle / \langle bk - \ell \rangle \rightarrow b^{-1}$; the quotient of the two logarithms tends to one. The difference operator for this ℓ is a bounded partial frequency shift followed by a diagonal multiplier tending to zero at both ends of $\mathbb{Z}$. Such a diagonal multiplier is a norm limit of finite-rank truncations. Summing over the finitely many ℓ proves (4.2). Taking powers preserves equality modulo compact operators and gives (4.3). $\square$

The Fredholm essential spectrum is invariant under compact perturbations. Together with theorem 3.4, theorem 4.1 already proves (2.5). This step alone does *not* exclude additional isolated eigenvalues outside the disk. The finite-frequency argument below does that work.

4.2 Finite bands and quotient tails

Lemma 4.2 (Invariance and strict descent). *For every $N \geq d$, F_N is invariant under $\mathcal{L}_{b,w}$. If a trigonometric polynomial has degree $p > d$, its image has degree at most $\lfloor (p + D)/b \rfloor < p$. Therefore the induced operator on F_N/F_d is nilpotent.*

Proof. A nonzero output frequency satisfies $|bk - j| \leq D$, so its absolute value is at most $(|j| + D)/b$. For $N \geq d$, write $D = (b - 1)d + e$ with $0 \leq e < b - 1$. Then $\lfloor (N + D)/b \rfloor \leq N$. If $p > d$, one has $D < (b - 1)p$, which gives strict descent. Iteration reaches F_d in finitely many steps. $\square$

Lemma 4.3 (Every exterior spectral point is finite-dimensional). *Under (P), if $|\lambda| > r_s$, then $\lambda \in \sigma(\mathcal{L}_{b,w}; H^{s,\beta})$ if and only if $\lambda \in \sigma(A_{b,w})$. The generalized eigenspaces at such λ are contained in F_d .*

Proof. Write $\tilde{L} = U_{s,\beta} \mathcal{L}_{b,w} U_{s,\beta}^{-1}$. Let Q_N be the orthogonal projection onto frequencies $|k| > N$. By (3.3), choose a positive integer n with

$$b^{-ns} \left\| \mathcal{L}_{b,w}^n \right\|_{2 \rightarrow 2} < |\lambda|^n.$$

Using (4.3) and $\|Q_N K_{s,\beta,n} Q_N\| \rightarrow 0$, choose $N \geq d$ so large that

$$\left\| Q_N \tilde{L}^n Q_N \right\| < |\lambda|^n. \quad (4.5)$$

Since F_N is invariant, the quotient operator is represented on its orthogonal complement by $T_N = Q_N \tilde{L} Q_N$, and $T_N^n = Q_N \tilde{L}^n Q_N$ on that complement. Hence $\lambda I - T_N$ is invertible: its spectral radius is smaller than $|\lambda|$.

Relative to $F_N \oplus F_N^\perp$, the operator is block upper triangular. The lower diagonal block is invertible after subtracting λI . Thus the full operator is invertible exactly when its finite block on F_N is invertible. By theorem 4.2, that finite block has the same nonzero eigenvalues as the block on F_d ; the remaining quotient is nilpotent. Also, if $(\tilde{L} - \lambda I)^j v = 0$, invertibility on the tail forces $v \in F_N$, and invertibility of the nilpotent quotient minus λI forces $v \in F_d$. Diagonal conjugation by $U_{s,\beta}$ preserves each finite band, completing the proof. $\square$

Completion of theorem 2.1. The essential disk follows from compact conjugacy. Every finite-matrix eigenvalue is an eigenvalue on the full space, since F_d is invariant. Theorem 4.3 excludes any other spectrum outside the disk and identifies the generalized eigenspaces. Near an exterior eigenvalue the quotient block remains invertible, so the finite-block resolvent also identifies its algebraic multiplicity and Jordan structure.

Finally, the Calkin image of $\mathcal{L}_{b,w}$ has spectrum $\overline{D}(0, r_s)$. The spectrum of its n th power is $\overline{D}(0, r_s^n)$ by spectral mapping. Its norm is at least its spectral radius, and the quotient norm is no larger than $\|\mathcal{L}_{b,w}^n - K\|$ for any compact K . This proves (2.7). $\square$

Remark 4.4 (Regularity-dependent, not contradictory spectra). An analytic Hardy space and a finite-order Sobolev space are different Banach spaces with different resolvents. The appearance of an essential disk here neither contradicts nor weakens the analytic finite-spectrum result in the repository. It quantifies precisely what analytic regularity had removed.

5 All even sine powers and integer interval regularity

5.1 A positive eigenfunction exists in a finite cone

For $m \geq 1$, let

$$w_m(x) = \sin^{2m}(\pi x), \quad c_{m,\ell} = \frac{(-1)^\ell}{4^m} \binom{2m}{m+\ell} \quad (|\ell| \leq m). \quad (5.1)$$

The coefficients are zero outside this interval. Denote the matrix (2.4) for w_m by $A_{b,m}$.

Proposition 5.1 (Automatic positivity for even sine masks). *For every $b \geq 2$ and $m \geq 1$, there exist a strictly positive real trigonometric polynomial h and a number $\rho_{b,m} > 0$ such that*

$$\mathcal{L}_{b,w_{2m}} h = \rho_{b,m} h. \quad (5.2)$$

The degree of h is at most $d_2 = \lfloor 2m/(b-1) \rfloor$, and

$$\rho_{b,m} = r(A_{b,2m}), \quad (5.3)$$

where r denotes spectral radius. In particular, $\rho_{b,m}$ is a positive algebraic number.

Proof. Let V be the real vector space of real-valued trigonometric polynomials of degree at most d_2 . Consider the set

$$\mathcal{K} = \left\{ f \in V : f \geq 0, \int_0^1 f = 1 \right\}.$$

It is nonempty and convex. It is compact because, for $f \in \mathcal{K}$, $|\hat{f}(k)| \leq \int f = 1$, and it is closed in the finite-dimensional space V . By the finite-band lemma, $\mathcal{L}_{b,w_{2m}}$ maps V into V . For

nonzero nonnegative f , the integral $\int w_{2m}f$ is strictly positive, since $w_{2m} > 0$ except at the integer point and f is a nonzero continuous function. Therefore

$$f \mapsto \frac{\mathcal{L}_{b,w_{2m}}f}{\int_0^1 \mathcal{L}_{b,w_{2m}}f}$$

is a continuous map from $\mathcal{K}$ to itself. The finite-dimensional Brouwer fixed-point theorem gives an eigenfunction $h \geq 0$ with mean one and eigenvalue $\rho > 0$.

It remains to prove strict positivity. A nonzero trigonometric polynomial of degree at most d_2 has at most $2d_2$ distinct zeros on $\mathbb{T}$: multiply its Laurent polynomial by the d_2 th power of its variable and use the ordinary polynomial zero bound. Choose n with $(b-1)b^{n-1} > 2d_2$. If $x \in (0, 1)$, all b^n inverse branches in $\mathcal{L}_{b,w_{2m}}^n h(x)$ have strictly positive product weights. If $x = 0$, at least the $(b-1)b^{n-1}$ preimages k/b^n with $b \nmid k$ have strictly positive product weights; none of their first n forward positions is an integer. There are therefore more positive-weight preimages than zeros of h . At least one summand is positive, at every x , so $\mathcal{L}_{b,w_{2m}}^n h(x) > 0$. Since this equals $\rho^n h(x)$, we obtain $h > 0$.

On the complexification of V , equip functions with the equivalent norm $\|f\|_h = \max_x |f(x)|/h(x)$. Positivity gives $|\mathcal{L}_{b,w_{2m}}f| \leq \mathcal{L}_{b,w_{2m}}|f| \leq \rho \|f\|_h h$. Hence its finite-dimensional spectral radius is at most ρ , and the eigenfunction gives equality. Its matrix has rational entries by (5.1), proving algebraicity. $\square$

Corollary 5.2 (Complete even-moment spectra). *For all integers $b \geq 2$, $m \geq 1$, and real s, β ,*

$$\begin{aligned} \sigma_{\text{ess}}(\mathcal{L}_{b,w_m}; H^{s,\beta}) &= \overline{D}(0, b^{-s} \sqrt{r(A_{b,2m})}), \\ \sigma(\mathcal{L}_{b,w_m}; H^{s,\beta}) &= \overline{D}(0, b^{-s} \sqrt{r(A_{b,2m})}) \cup \sigma(A_{b,m}). \end{aligned} \quad (5.4)$$

An eigenvalue $\lambda \neq 0$ of $A_{b,m}$ lies outside the essential disk exactly when

$$s > \log_b \left(\frac{\sqrt{r(A_{b,2m})}}{|\lambda|} \right). \quad (5.5)$$

This gives a finite algebraic procedure for the complete spectral set at any specified regularity. Numerical diagonalization is not part of the proof. Exact characteristic polynomials or certified algebraic root isolation may be used to turn the formulas into explicit numbers.

5.2 Unmatched interval jets do not add nonzero spectrum

Let $H^r(0, 1)$ be the usual interval Sobolev space at an integer order $r \geq 0$. When $r \geq 1$, its periodic subspace is characterized by the r matching conditions

$$f^{(j)}(1) = f^{(j)}(0), \quad 0 \leq j < r. \quad (5.6)$$

It is naturally isomorphic to $H^r(\mathbb{T})$, with equivalent norms. Define $\delta_j(f) = f^{(j)}(1) - f^{(j)}(0)$.

Lemma 5.3 (Exact boundary-jet action). *For a smooth periodic mask w and $0 \leq j < r$,*

$$\delta_j(\mathcal{L}_{b,w}f) = b^{-j-1} \sum_{\ell=0}^j \binom{j}{\ell} w^{(j-\ell)}(0) \delta_\ell(f). \quad (5.7)$$

For $w = w_m$, the induced operator on the quotient by the periodic subspace is nilpotent, of nilpotency index at most $\lceil r/(2m) \rceil$ when $r > 0$.

Proof. Differentiate the branch formula j times. The difference between its values at 1 and 0 telescopes over the b branch endpoints, leaving $b^{-j-1}[(wf)^{(j)}(1) - (wf)^{(j)}(0)]$. Leibniz's rule and periodicity of all derivatives of w give (5.7). The identity holds first for smooth functions and then by Sobolev trace continuity.

For w_m , the derivatives of orders $0, \dots, 2m-1$ vanish at zero. Thus the output jet of index j depends only on input jets of indices at most $j-2m$. Iterating $\lceil r/(2m) \rceil$ times annihilates all r quotient coordinates. These coordinates describe the full quotient: the jet map is onto by elementary Hermite interpolation. $\square$

Theorem 5.4 (Integer-order interval spectrum). *For integers $r \geq 0$, $b \geq 2$, $m \geq 1$, the complete spectrum and the Fredholm essential spectrum of $\mathcal{L}_{b,w_m}$ on $H^r(0,1)$ are exactly those in (5.4) with $s = r$ and $\beta = 0$. Every generalized eigenvector at a nonzero eigenvalue has matching endpoint jets and belongs to the periodic subspace.*

Proof. For $r = 0$, interval and periodic L^2 are the same Hilbert space up to identification of endpoints of measure zero. For $r > 0$, the periodic subspace is invariant and has finite codimension r . Choose any bounded finite-dimensional complement. The resulting block upper triangular operator has the periodic operator on one diagonal block and the nilpotent jet operator on the other. For $\lambda \neq 0$, the quotient block minus λI is invertible. The full operator is therefore invertible exactly when the periodic block is invertible. The point zero already belongs to the periodic essential disk. A finite-dimensional block and the off-diagonal finite-rank block do not change Fredholm essential spectrum. Finally, projecting a nonzero λ -generalized eigenvector to the nilpotent quotient gives zero, which proves the matching assertion. $\square$

This argument does not justify the same statement on fractional interval Sobolev spaces. Trace conditions change at half-integer orders, and the critical logarithmic spaces require separate endpoint analysis. That extension is retained as a research question rather than silently assumed.

6 The quadratic dyadic operator: exact thresholds

6.1 The two finite matrices

For $b = 2, m = 1$, the terminal Fourier band is F_1 . In the basis (e_{-1}, e_0, e_1) ,

$$A_{2,1} = \begin{pmatrix} -1/4 & 0 & 0 \\ -1/4 & 1/2 & -1/4 \\ 0 & 0 & -1/4 \end{pmatrix}, \quad \det(tI - A_{2,1}) = (t - 1/2)(t + 1/4)^2. \quad (6.1)$$

The eigenspaces are generated by

$$1, \quad v_+ = e_1 + \frac{1}{3}, \quad v_- = e_{-1} + \frac{1}{3}, \quad (6.2)$$

with respective eigenvalues $1/2, -1/4, -1/4$. Both values are semisimple in this finite block.

The squared mask is $w_2 = \sin^4(\pi x)$. In the basis $(e_{-2}, e_{-1}, e_0, e_1, e_2)$ its matrix is

$$A_{2,2} = \begin{pmatrix} 1/16 & 0 & 0 & 0 & 0 \\ 3/8 & -1/4 & 1/16 & 0 & 0 \\ 1/16 & -1/4 & 3/8 & -1/4 & 1/16 \\ 0 & 0 & 1/16 & -1/4 & 3/8 \\ 0 & 0 & 0 & 0 & 1/16 \end{pmatrix}. \quad (6.3)$$

Its characteristic polynomial is

$$(t + 1/4)(t - 1/16)^2(t^2 - t/8 - 1/16). \quad (6.4)$$

In particular,

$$\rho = \frac{1 + \sqrt{17}}{16}, \quad h(x) = 1 + \kappa \cos(2\pi x), \quad \kappa = \frac{5 - \sqrt{17}}{4} \quad (6.5)$$

satisfy $\mathcal{L}_{2,w_2}h = \rho h$. Since $0 < \kappa < 1$, this also gives a direct check of the strict positivity hypothesis without the cone argument. The exact identities in this subsection are elementary matrix computations; their role is to make the general theorem explicit, not to claim these finite eigenvalues as newly discovered.

Theorem 6.1 (Dyadic regularity transition). *For every real s, β , the quadratic dyadic operator (1.1) has the spectrum (1.3). Its eigenvalue $1/2$ is isolated if and only if $s > s_0$, and $-1/4$ is isolated if and only if $s > s_1$. When isolated, their algebraic multiplicities are respectively one and two, and both are semisimple. The same spectral sets hold on $H^r(0, 1)$ at integer $r \geq 0$.*

Proof. Use theorems 2.1 and 5.4 and (6.1)–(6.5). Solving $\gamma 2^{-s} < 1/2$ and $\gamma 2^{-s} < 1/4$ gives the two strict thresholds. At equality, the eigenvalue is on the boundary of a disk contained in the spectrum, and is therefore not isolated. $\square$

For orientation, the constants have the approximate values

$$\gamma \approx 0.5658569621, \quad s_0 \approx 0.1785093184, \quad s_1 \approx 1.1785093184. \quad (6.6)$$

These decimals are explanatory approximations; the exact radicals and logarithms, not rounded values, define the thresholds.

Space	Essential disk radius	Isolated displayed eigenvalues
L^2	γ	none
H^1	$\gamma/2$	$1/2$ only
$H^{s_1, \beta}$	$1/4$	$1/2$ only
H^2	$\gamma/4$	$1/2$ and $-1/4$

The table concerns isolation, not existence: the polynomial eigenfunctions (6.2) belong to all these spaces. In particular, one full derivative is not enough to isolate the alternating eigenvalue.

7 Dual distributions and an exact Fourier cutoff

7.1 Two familiar recurrences with different spectral roles

Define real sequences $(\eta_k)_{k \geq 0}$ and $(a_k)_{k \geq 0}$ by

$$\eta_0 = 1, \quad \eta_1 = -\frac{1}{3}, \quad \eta_{2k} = \eta_k, \quad \eta_{2k+1} = -\frac{1}{2}(\eta_k + \eta_{k+1}), \quad (7.1)$$

$$a_0 = 0, \quad a_1 = 1, \quad a_{2k} = -2a_k, \quad a_{2k+1} = a_k + a_{k+1}. \quad (7.2)$$

The potentially self-referential odd equations at $k = 0$ are consistent with these initial values. For subsequent indices they give an unambiguous recursion. The sequence η is the balanced Thue–Morse pair-correlation sequence; $a_k = B_k(-2)$ in the standard Stern-polynomial convention [5, 6]. Neither sequence is introduced here as a new integer or rational sequence.

For a trigonometric polynomial f , define

$$\begin{aligned}\mu(f) &= \sum_{k \in \mathbb{Z}} \eta_{|k|} \widehat{f}(k), \\ D_+(f) &= \sum_{k \geq 1} a_k \widehat{f}(k), \quad D_-(f) = \sum_{k \leq -1} a_{|k|} \widehat{f}(k).\end{aligned}\tag{7.3}$$

These formulas define distributions even before their sharp Sobolev continuity thresholds are known: $|\eta_k| \leq 1$ by induction, and $|a_k| \leq k$ for $k \geq 1$, again by induction using (7.2).

Proposition 7.1 (Dual eigenrelations and biorthogonality). *On trigonometric polynomials,*

$$\mu \mathcal{L} = \tfrac{1}{2} \mu, \quad D_+ \mathcal{L} = -\tfrac{1}{4} D_+, \quad D_- \mathcal{L} = -\tfrac{1}{4} D_-.\tag{7.4}$$

Moreover,

$$\mu(1) = 1, \quad \mu(v_{\pm}) = 0, \quad D_{\pm}(1) = 0, \quad D_+(v_+) = D_-(v_-) = 1, \quad D_+(v_-) = D_-(v_+) = 0.\tag{7.5}$$

Consequently the algebraic operators

$$P_0 f = \mu(f) 1, \quad P_1 f = D_+(f) v_+ + D_-(f) v_- \tag{7.6}$$

are disjoint idempotents commuting with $\mathcal{L}$ on polynomials.

Proof. The Fourier action is

$$\mathcal{L} e_{2k} = \tfrac{1}{2} e_k, \quad \mathcal{L} e_{2k+1} = -\tfrac{1}{4} (e_k + e_{k+1}). \tag{7.7}$$

For μ , its evaluation against these identities is exactly (7.1). For D_+ on positive indices, it is exactly (7.2); the mode at zero is annihilated, and negative modes cannot produce a positive output frequency. The D_- argument is symmetric. Direct evaluation on $1, v_+, v_-$ gives (7.5) and then the projection identities. $\square$

7.2 The finite Riesz product and its stable coefficient formula

Write

$$p_n(x) = 2^n W_n(x) = \prod_{j=0}^{n-1} (1 - \cos(2\pi 2^j x)), \quad q_n(k) = \widehat{p}_n(k). \tag{7.8}$$

The polynomial p_n is nonnegative, has degree $2^n - 1$, and has mean one. Indeed, $p_0 = 1$ and the identity $p_{n+1}(x) = (1 - \cos(2\pi x)) p_n(2x)$ gives

$$q_{n+1}(2k) = q_n(k), \quad q_{n+1}(2k+1) = -\tfrac{1}{2} (q_n(k) + q_n(k+1)). \tag{7.9}$$

In particular $q_n(0) = 1$ and $q_n(-k) = q_n(k)$.

Theorem 7.2 (Exact cutoff identity). *For every $n \geq 0$ and $0 \leq k \leq 2^n$,*

$$q_n(k) = \eta_k + \frac{1}{3} (-1/2)^n a_k. \tag{7.10}$$

For $|k| \geq 2^n$, $q_n(k) = 0$. At $k = 2^n$ both statements agree.

Proof. At $n = 0$, the required indices are $k = 0, 1$. The right side is respectively 1 and $-1/3 + 1/3 = 0$, as required. Assume the formula at level n . For an even index $2k \leq 2^{n+1}$, combine $q_{n+1}(2k) = q_n(k)$ with $\eta_{2k} = \eta_k$ and $a_{2k} = -2a_k$; the correction factor satisfies $(-1/2)^{n+1}(-2) = (-1/2)^n$. For an odd index $2k+1 \leq 2^{n+1}$, the two indices $k, k+1$ lie in the previous valid range. Substitution in (7.9), followed by the two sequence recurrences, gives the formula. The support statement follows from the degree of p_n . At its boundary, $\eta_{2^n} = -1/3$ and $a_{2^n} = (-2)^n$, so the right side is zero. $\square$

As a consequence, $q_n(k) \rightarrow \eta_{|k|}$ for every fixed integer k . The probability measures $p_n(x) dx$ therefore converge weakly to a probability measure whose integral against every trigonometric polynomial is μ . To see this without any measure-theoretic identification theorem, use compactness of probability measures on the compact circle to obtain subsequential limits, then use convergence on all characters and density of trigonometric polynomials in $C(\mathbb{T})$ to prove uniqueness. Thus the notation μ can denote both the leading distribution and this measure. No singularity assertion is needed for any proof below.

7.3 The moment remainder is exactly a Fourier tail

For a polynomial f , change of variables in the transfer formula gives

$$I_n(f) = \int_0^1 \mathcal{L}^n f(x) dx. \quad (7.11)$$

For example, induction follows from $\int (\mathcal{L}g)h = \int wg(h \circ T_2)$.

Corollary 7.3 (Exact two-term identity with a vanishing low-frequency remainder). *On trigonometric polynomials, (1.5) holds with*

$$R_n(f) = - \sum_{|k| > 2^n} \left(2^{-n} \eta_{|k|} + \frac{(-1/4)^n}{3} a_{|k|} \right) \hat{f}(k). \quad (7.12)$$

In particular, R_n annihilates every polynomial of degree at most 2^n . Whenever μ, D_+, D_- are continuous on $H^{s,\beta}$, the identities extend to that whole space and the tail pairing converges absolutely by Cauchy–Schwarz.

Proof. Since q_n is even, integration gives $I_n(f) = 2^{-n} \sum_k q_n(k) \hat{f}(k)$. Apply theorem 7.2 on $|k| \leq 2^n$, and use its support assertion outside that interval. The coefficient identity $2^{-n}(-1/2)^n = (-1/4)^n$ gives the result. Extension follows by density and continuity; I_n is a finite Fourier pairing on every $H^{s,\beta}$. $\square$

Example 7.4 (A single frequency and the nonuniformity of the cutoff). For $f = e_{2^j}$ and $n \geq j$, the exact expression is

$$I_n(e_{2^j}) = -\frac{2^{-n}}{3} + \frac{(-1/4)^n (-2)^j}{3}.$$

For $n < j$, the moment is zero because its frequency lies beyond the support of p_n . At $n = j$, the displayed two terms cancel exactly. This illustrates why a fixed-frequency asymptotic is not by itself a uniform statement over a Sobolev unit ball: frequencies can move with n .

8 Quadratic energies and sharp continuity thresholds

8.1 Four finite-state recurrences

Define the prefix energies and adjacent correlations

$$\begin{aligned} E_n &= \sum_{k=0}^{2^n-1} a_k^2, & C_n &= \sum_{k=0}^{2^n-1} a_k a_{k+1}, \\ F_n &= \sum_{k=0}^{2^n-1} \eta_k^2, & G_n &= \sum_{k=0}^{2^n-1} \eta_k \eta_{k+1}. \end{aligned} \quad (8.1)$$

The letter F_n in this section denotes a scalar energy, not the finite Fourier space F_N ; the indices and context distinguish them.

Lemma 8.1 (Exact energy recurrences). *The initial values are $(E_0, C_0) = (0, 0)$ and $(F_0, G_0) = (1, -1/3)$. For every $n \geq 0$,*

$$E_{n+1} = 6E_n + 2C_n + 4^n, \quad C_{n+1} = -4E_n - 4C_n - 2 \cdot 4^n, \quad (8.2)$$

$$F_{n+1} = \frac{3}{2}F_n + \frac{1}{2}G_n - \frac{2}{9}, \quad G_{n+1} = -F_n - G_n + \frac{4}{9}. \quad (8.3)$$

Proof. Put $M = 2^n$. Splitting E_{n+1} into even and odd indices gives

$$E_{n+1} = \sum_{k=0}^{M-1} [4a_k^2 + (a_k + a_{k+1})^2].$$

Since $a_0 = 0$ and $a_M^2 = 4^n$, the shifted sum $\sum_{k=0}^{M-1} a_{k+1}^2$ equals $E_n + 4^n$. This proves the first recurrence. The adjacent products in an even-odd pair sum to $-2(a_k + a_{k+1})^2$, proving the second.

For η , the same split uses $\eta_{2k} = \eta_k$ and $\eta_{2k+1} = -(\eta_k + \eta_{k+1})/2$. The shifted square sum is now $F_n - 1 + 1/9 = F_n - 8/9$, since $\eta_M^2 = 1/9$. Thus

$$F_{n+1} = F_n + \frac{1}{4}(2F_n + 2G_n - 8/9), \quad G_{n+1} = -\frac{1}{2}(2F_n + 2G_n - 8/9),$$

which is (8.3). $\square$

8.2 Closed forms and dyadic annuli

Let $u = \sqrt{17}$, $\Lambda_{\pm} = 1 \pm u$, and $\lambda_{\pm} = (1 \pm u)/4$.

Proposition 8.2 (Closed quadratic-energy formulas). *For every $n \geq 0$,*

$$E_n = \frac{5+u}{4u}\Lambda_+^n + \frac{u-5}{4u}\Lambda_-^n - \frac{1}{2}4^n, \quad (8.4)$$

$$F_n = \frac{19+5u}{18u}\lambda_+^n + \frac{5u-19}{18u}\lambda_-^n + \frac{4}{9}. \quad (8.5)$$

Consequently, as $j \rightarrow \infty$,

$$\sum_{k=2^j}^{2^{j+1}-1} a_k^2 \sim c_a \Lambda_+^j, \quad c_a = \frac{5+u}{4}, \quad (8.6)$$

$$\sum_{k=2^j}^{2^{j+1}-1} \eta_k^2 \sim c_{\eta} \lambda_+^j, \quad c_{\eta} = \frac{19+5u}{18u} \frac{u-3}{4} > 0. \quad (8.7)$$

Proof. The homogeneous matrix in (8.2) is $\begin{pmatrix} 6 & 2 \\ -4 & -4 \end{pmatrix}$, with eigenvalues $1 \pm \sqrt{17}$. A particular solution has $E_n = -4^n/2$. Solving for the two homogeneous coefficients from $E_0 = 0$ and $E_1 = 1$ gives (8.4). The homogeneous matrix in (8.3) has eigenvalues $(1 \pm \sqrt{17})/4$, and a stationary solution has $F_n = 4/9$. Using $F_0 = 1$, $F_1 = 10/9$ gives (8.5). These computations can alternatively be checked by substituting into the recurrences and initial values, which also proves them by induction.

For E_n , the growth rate $\Lambda_+ = 1 + u$ is strictly larger than 4 and $|\Lambda_-|$. For F_n , $\lambda_+ > 1$ and $|\lambda_-| < 1$. Taking differences between consecutive prefix energies therefore gives the annulus equivalents and the stated positive constants. $\square$

Notice the exact relations

$$\Lambda_+ = 2^{2s_1}, \quad \lambda_+ = 2^{2s_0}, \quad \Lambda_+ = 4\lambda_+. \quad (8.8)$$

Thus the same two thresholds that govern spectral isolation also govern the growth of the dual coefficient energies. This agreement is derived, not assumed from a general regularity principle.

8.3 The Hilbert dual criterion

Lemma 8.3 (Coefficient criterion for a continuous functional). *Let $T(f) = \sum_k t_k \hat{f}(k)$ on polynomials. Then T extends continuously to $H^{s,\beta}$ if and only if*

$$\sum_{k \in \mathbb{Z}} |t_k|^2 \langle k \rangle^{-2s} [\log(e + |k|)]^{-2\beta} < \infty. \quad (8.9)$$

When finite, the sum is exactly $\|T\|_{(H^{s,\beta})^}^2$.*

Proof. Cauchy–Schwarz proves sufficiency and the upper norm bound. For necessity and equality, use finite-support coefficients proportional to $\bar{t}_k q_{s,\beta}(k)^{-2}$ and normalize their Sobolev norm. The resulting functional values are the square roots of the finite partial sums in (8.9). Let the support increase. $\square$

Theorem 8.4 (Sharp leading and subleading regularity thresholds). *The functional μ extends continuously to $H^{s,\beta}$ exactly when*

$$s > s_0 \quad \text{or} \quad s = s_0 \text{ and } \beta > 1/2. \quad (8.10)$$

Each of D_+ and D_- , and also their sum, extends continuously exactly when

$$s > s_1 \quad \text{or} \quad s = s_1 \text{ and } \beta > 1/2. \quad (8.11)$$

In their stated ranges, P_0 and P_1 are bounded commuting projections. When the corresponding eigenvalue is isolated, these are its Riesz projections.

Proof. On the dyadic block $2^j \leq |k| < 2^{j+1}$, the reciprocal squared Sobolev weight is comparable, with constants independent of $j \geq 1$, to $2^{-2sj}(1+j)^{-2\beta}$. The annulus formulas show that the positive frequency contributions in (8.9) are comparable to

$$2^{2(s_0-s)j}(1+j)^{-2\beta} \quad \text{for } \mu, \quad 2^{2(s_1-s)j}(1+j)^{-2\beta} \quad \text{for } D_+. \quad (8.12)$$

Negative frequencies give an identical test. The geometric series converges above the stated exponent, diverges below it, and at equality reduces to the power-series test $\sum j^{-2\beta} < \infty$. For $D_+ + D_-$, the supports are disjoint, so there is no cancellation in the squared dual norm. This proves necessity as well as sufficiency.

Bounded projection identities follow from theorem 7.1 and density. When an eigenvalue is outside the essential disk, its generalized eigenspace is exactly its semisimple finite Fourier eigenspace by theorem 2.1. On every polynomial, sufficiently many iterations descend into F_1 , where (7.5) gives the finite spectral projection. Thus the displayed projection agrees on a dense subspace with the Riesz projection, and hence everywhere. $\square$

Remark 8.5 (Continuous embedded projections). At $s = s_1$, $\beta > 1/2$, the finite-rank operator P_1 is bounded and commutes with $\mathcal{L}$, but $-1/4$ is not isolated. It is therefore not a Riesz projection defined by a contour separating that point from the rest of the spectrum. A bounded projection onto a finite-dimensional invariant summand is compatible with other spectrum accumulating at the same eigenvalue. The corresponding observation applies to P_0 at $s = s_0$, $\beta > 1/2$.

9 Sharp scalar remainders, including the critical endpoint

9.1 Weighted tails of the two coefficient sequences

For $z = (z_k)_{k \geq 1}$, put

$$\mathcal{T}_z(N; s, \beta)^2 = \sum_{k > N} |z_k|^2 \langle k \rangle^{-2s} [\log(e + k)]^{-2\beta}. \quad (9.1)$$

We use $X_n \asymp Y_n$ to mean that there exist positive constants, independent of sufficiently large n , bounding X_n/Y_n above and below. Constants may depend on the fixed displayed parameters.

Lemma 9.1 (Two-sided tail estimates). *Let $\sigma_a = s_1$ and $\sigma_\eta = s_0$. For $z = a$ or $z = \eta$, if $s > \sigma_z$, then*

$$\mathcal{T}_z(2^n; s, \beta) \asymp_{s, \beta} 2^{(\sigma_z - s)n} (1 + n)^{-\beta}. \quad (9.2)$$

If $s = \sigma_z$ and $\beta > 1/2$, then

$$\mathcal{T}_z(2^n; \sigma_z, \beta) \asymp_\beta (1 + n)^{1/2 - \beta}. \quad (9.3)$$

The analogous norm restricted to one block $2^j \leq k < 2^{j+1}$ is comparable to $2^{(\sigma_z - s)j} (1 + j)^{-\beta}$.

Proof. Apply the annulus estimates in theorem 8.2 and the comparability of the weights on each dyadic block. The squared tail is comparable to

$$\sum_{j \geq n} 2^{2(\sigma_z - s)j} (1 + j)^{-2\beta}.$$

Deleting the single boundary term $k = 2^n$ does not change the comparison. For a , its square is $4^n = o(2^{2s_1 n})$; for η , it is $1/9$ while the annular energy grows like $2^{2s_0 n}$. When $s > \sigma_z$, a decaying geometric factor makes the sum comparable to its initial term, even for negative β . At $s = \sigma_z$, the integral test makes the sum comparable to $(1 + n)^{1 - 2\beta}$. Taking square roots proves all the assertions. $\square$

9.2 The exact rate above critical regularity

Theorem 9.2 (Optimal supercritical scalar remainder). *For every $s > s_1$ and every real β , the expansion (1.5) defines continuous functionals on $H^{s, \beta}$, and*

$$\|R_n\|_{(H^{s, \beta})^*} \asymp_{s, \beta} (\gamma 2^{-s})^n (1 + n)^{-\beta}. \quad (9.4)$$

The upper estimate is uniform for f in the Sobolev unit ball, and its exponential rate and displayed power of n cannot be improved uniformly. In particular, $\|R_n\| = o(4^{-n})$ in this range.

Proof. Continuity follows from theorem 8.4. By the exact tail formula and the dual norm criterion, $\|R_n\|$ is the weighted ℓ^2 norm of its coefficient sequence. The triangle inequality and theorem 9.1 give an upper bound by a fixed constant times

$$\begin{aligned} & 2^{-n} 2^{(s_0 - s)n} (1 + n)^{-\beta} + 4^{-n} 2^{(s_1 - s)n} (1 + n)^{-\beta} \\ &= 2(\gamma 2^{-s})^n (1 + n)^{-\beta}, \end{aligned}$$

where the harmless factor $1/3$ in the second coefficient is absorbed into the constant. The equality of the two exponential rates uses $2^{s_0}/2 = 2^{s_1}/4 = \gamma$.

For the lower bound, cancellation between the two coefficient sequences must be excluded; separate upper estimates would not suffice. Fix a large integer $L \geq 1$ and restrict to the block $2^{n+L} \leq k < 2^{n+L+1}$. Let $A_{n,L}$ and $B_{n,L}$ be the weighted ℓ^2 norms of, respectively, the Stern term and the correlation term on this block. The one-block estimates give

$$\begin{aligned} A_{n,L} &\asymp 4^{-n} 2^{(s_1 - s)(n+L)} (1 + n + L)^{-\beta}, \\ B_{n,L} &\asymp 2^{-n} 2^{(s_0 - s)(n+L)} (1 + n + L)^{-\beta}. \end{aligned}$$

The implicit comparison constants can be chosen independently of n, L when $n + L$ is sufficiently large. Hence $A_{n,L}/B_{n,L} \geq c_{s, \beta} 2^L$ for some positive constant. Choose and fix L so that this lower bound exceeds two. The reverse triangle inequality then makes the norm of the sum at least $A_{n,L} - B_{n,L} \geq A_{n,L}/2$. With this fixed L , that lower bound is a positive constant times the right side of (9.4). The dual norm criterion identifies this coefficient norm with the exact supremum over the unit ball, proving optimality. Finally, $\gamma 2^{-s} < 1/4$ when $s > s_1$. $\square$

9.3 A sharp logarithmic rescue at the spectral boundary

Theorem 9.3 (Critical two-term scalar asymptotic). *At $s = s_1$ and $\beta > 1/2$, the two-term expansion (1.5) is valid on all of $H^{s_1, \beta}$ and*

$$\|R_n\|_{(H^{s_1, \beta})^*} \asymp_\beta 4^{-n}(1+n)^{1/2-\beta}. \quad (9.5)$$

Equivalently,

$$2^n I_n(f) = \mu(f) + \frac{(-1/2)^n}{3}(D_+ + D_-)(f) + O_\beta\left(2^{-n}(1+n)^{1/2-\beta} \|f\|_{s_1, \beta}\right). \quad (9.6)$$

The error is uniformly $o(2^{-n})$ after this normalization. The power $(1+n)^{1/2-\beta}$ is sharp in the dual norm.

Proof. The coefficient functionals are bounded by theorem 8.4. The weighted norm of the Stern part of the exact tail is comparable to $4^{-n}(1+n)^{1/2-\beta}$ by the critical tail estimate. The correlation part, which is one derivative above its own critical index, has norm comparable to

$$2^{-n}2^{(s_0-s_1)n}(1+n)^{-\beta} = 4^{-n}(1+n)^{-\beta}.$$

The second quantity is smaller than the first by a factor comparable to $(1+n)^{-1/2}$. Thus both the triangle and reverse triangle inequalities give (9.5) for sufficiently large n . The two frequency halves are disjoint and symmetric, contributing only a fixed factor of $\sqrt{2}$. Multiplication by 2^n gives (9.6); since $\beta > 1/2$, its extra power tends to zero. $\square$

Corollary 9.4 (Necessity of the logarithmic threshold). *On $H^{s_1, \beta}$ with $\beta \leq 1/2$, there do not exist bounded linear functionals M, S and continuous remainders ε_n satisfying*

$$I_n = 2^{-n}M + (-1/4)^n S + \varepsilon_n, \quad \|\varepsilon_n\| = o(4^{-n}).$$

The same impossibility holds on $H^{s, \beta}$ for every $s < s_1$.

Proof. For a fixed polynomial, the remainder vanishes once 2^n exceeds its degree. Thus the leading and subleading coefficients are uniquely fixed by limits: the leading one is $\lim_n 2^n I_n(f) = \mu(f)$, and after its subtraction the rescaled subleading coefficient is $(D_+ + D_-)(f)/3$. Any continuous coefficient functional agreeing with those polynomial limits would extend $D_+ + D_-$. The necessity part of theorem 8.4 says that such an extension does not exist in the stated range. $\square$

This is an obstruction to a *uniform continuous-coefficient theorem* on the whole Hilbert space. It does not say that a particular smoother amplitude, or a polynomial, fails to have a two-term expansion. Nor does it rule out different formulations using unbounded coefficient maps on proper domains.

10 Scalar asymptotics versus operator asymptotics

10.1 The supercritical operator expansion

Proposition 10.1 (Operator remainder and its exact exponential scale). *For $s > s_1$ and any real β , put $Q = \mathcal{L}(I - P_0 - P_1)$ on $H^{s, \beta}$. For $n \geq 1$,*

$$\mathcal{L}^n = 2^{-n}P_0 + (-1/4)^n P_1 + Q^n, \quad \sigma(Q) = \overline{D}(0, \gamma 2^{-s}). \quad (10.1)$$

Consequently,

$$\lim_{n \rightarrow \infty} \|Q^n\|^{1/n} = \gamma 2^{-s}, \quad \|Q^n\| \leq C_\varepsilon (\gamma 2^{-s} + \varepsilon)^n \quad (\varepsilon > 0). \quad (10.2)$$

Also $\|Q^n\| \geq (\gamma 2^{-s})^n$ for every $n \geq 1$.

Proof. The bounded disjoint projections commute with $\mathcal{L}$ and have the prescribed eigenvalues on their ranges. They are the Riesz projections for the two isolated spectral values. Removing their summands leaves exactly the essential disk; adding zero on their finite-dimensional ranges does not enlarge it. This proves (10.1). The spectral radius formula proves (10.2). The lower bound follows either from that spectral set and spectral mapping at each n , or from (2.7) after subtracting the two finite-rank terms. $\square$

The general spectral argument yields an arbitrarily small exponential slack ε in an upper operator-norm bound. The scalar moment calculation in theorem 9.2 is stronger for its specific observable: it removes that slack and supplies the exact logarithmic power. One must not promote that scalar estimate to the full operator norm without a separate proof.

10.2 At critical regularity the two notions separate

Theorem 10.2 (A scalar little-oh law without an operator little-oh law). *On $H^{s_1, \beta}$ with $\beta > 1/2$, the operators P_0, P_1 remain bounded and the algebraic power identity in (10.1) remains valid. Nevertheless,*

$$\|\mathcal{L}^n - 2^{-n}P_0 - (-1/4)^n P_1\| \geq 4^{-n} \quad (n \geq 1). \quad (10.3)$$

For the integration functional $\ell_0(f) = \int_0^1 f(x) dx$,

$$\|\ell_0(\mathcal{L}^n - 2^{-n}P_0 - (-1/4)^n P_1)\| \asymp_\beta 4^{-n}(1+n)^{1/2-\beta} = o(4^{-n}). \quad (10.4)$$

Proof. The projection identities were established algebraically and extend by boundedness, so the power decomposition remains true without an isolation assumption. Both subtracted terms have finite rank. The essential disk has radius $1/4$ at $s = s_1$, and the compact obstruction in theorem 2.1 proves (10.3). On the other hand, $\ell_0(1) = 1$ and $\ell_0(v_+) = \ell_0(v_-) = 1/3$. Therefore the functional on the left of (10.4) is exactly R_n . Apply theorem 9.3. $\square$

The source of the improvement is visible in (7.12): integration after n transfer steps annihilates the remainder on every frequency up to 2^n . The operator itself has no comparable finite-rank annihilation of its full essential component. The critical scalar law is therefore not an exception to the spectral obstruction, but a different kind of asymptotic statement.

10.3 How the result relates to Rvachev Fourier shells

The repository uses the entire product

$$\Phi(t) = \prod_{j=0}^{\infty} \operatorname{sinc}\left(\frac{\pi t}{2^j}\right), \quad \operatorname{sinc} z = \frac{\sin z}{z}, \quad \operatorname{sinc} 0 = 1. \quad (10.5)$$

This is its Fourier-product normalization for the Rvachev up-function [2, 7]. Its finite-scale identity, obtained simply by separating the first n factors, is

$$\Phi(2^n x) = \Phi(x) \prod_{j=1}^n \frac{\sin(\pi 2^j x)}{\pi 2^j x}. \quad (10.6)$$

For real $x \neq 0$, squaring and clearing the denominators gives

$$(\pi x)^{2n} 2^{n(n+1)} |\Phi(2^n x)|^2 = |\Phi(x)|^2 W_n(2x). \quad (10.7)$$

This exhibits directly the dyadic sine-product moments underlying shell analysis. The repository's precise RMS normalization packages the remaining factors into an analytic amplitude; that analytic application and its certified constants are credited to the antecedent manuscript, not recomputed or claimed as new here.

The new information is what remains true after weakening regularity. For periodic amplitudes, theorems 9.2 and 9.3 give sharp uniform two-term laws. For nonperiodic amplitudes on an interval, theorem 5.4 gives the exact integer-order spectral picture: two Sobolev derivatives suffice to isolate both displayed dyadic eigenvalues, whereas one does not. A nonperiodic analytic amplitude must not be declared to lie in every periodic Sobolev space; mismatched endpoint values or jets can prevent that. The fractional periodic endpoint theorem consequently requires its periodic hypothesis when applied to an amplitude.

11 Verification, implementation, and formalization boundaries

11.1 What the accompanying computation checks

The delivered program `verify.py` uses only Python's standard library. It constructs η_k and a_k exactly, forms the finite Riesz products by independent Laurent-polynomial multiplication, and compares their coefficients with (7.10). It checks the energy recurrences against direct finite sums and the closed forms in the exact field $\mathbb{Q}(\sqrt{17})$, represented as pairs of rational numbers. It also checks the dual eigenrelations, finite-band invariance and strict descent, the two displayed characteristic polynomials, and the explicit positive square-mask eigenfunction.

The executed default run, through dyadic level 12, returned **PASS**. Its recorded checks are:

Checked identity or property	Number of checks
Exact cutoff coefficients, including both signs	16,395
Dual eigenfunctional identities	24,579
Finite-band invariance and descent cases	89,014
Direct energy recurrences	52
Closed energy formulas in $\mathbb{Q}(\sqrt{17})$	26
Positive eigenfunction coordinates	5
Characteristic polynomial identities	2

The frequency checks reach $|k| = 4096$; the band tests cover $b = 2, \dots, 8$ and $m = 1, \dots, 6$. These finite ranges are computational scope statements, not substitutes for quantifiers in the theorems.

Run, from the extracted directory,

```
python verify.py --output verification.json
```

The output file includes the exact energy data, arithmetic type, Python version, and check counts. Explicit exceptions enforce the checks even if Python assertions are disabled. No numerical eigensolver, floating-point comparison, or unvalidated quadrature enters a passed mathematical check. The approximate decimals in (6.6) are not used by the verification program.

11.2 What is proved by analysis, not by sampling

The infinite-dimensional spectrum is proved through the coisometry similarity, construction of infinitely many defect eigenvectors, compact conjugacy, and the quotient-tail argument. The endpoint norm equivalents are proved through exact energy recurrences, dyadic comparisons, and the Hilbert dual criterion. No finite list of Fourier coefficients could establish those conclusions by itself.

The external functional-analytic inputs are standard: invariance of the Fredholm essential spectrum under compact perturbations, its description through the Calkin algebra, spectral mapping and the spectral radius formula, finite-dimensional Brouwer fixed point, and the integer-order Sobolev trace theorem. The special operators and estimates needed here are proved in the text. The dependency structure is:

Conclusion	Main ingredients
Exact L^2 disk	Square-mask identity, strict positivity, infinite branch defect
Sobolev disk	Fourier weight ratio and compactness of vanishing diagonals
Complete outlier classification	Finite-band descent and invertible quotient tails
Even sine family	Finite cone, zero count, and Brouwer fixed point
Interval theorem	Exact boundary jets and finite-codimensional nilpotent quotient
Critical scalar law	Cutoff identity, exact energies, and weighted dual norms

11.3 A realistic route to formalization

A first formalization layer should encode the two sequences over $\mathbb{Q}$ and $\mathbb{Z}$, prove their recurrences, and formalize the finite coefficient identity. These statements require only finite sums, parity decompositions, and induction. The quadratic-energy formulas can be formalized either in a quadratic field or by verifying their second-order recurrences without introducing radicals until the final corollary.

A second layer should prove the dyadic weighted ℓ^2 criterion and its geometric and critical power-series tests. That layer already yields the sharp continuity thresholds and scalar remainder results independently of Calkin-algebra infrastructure. The operator-spectrum layer then requires bounded Fourier multipliers, compact operators, quotient spectra, and Fredholm theory. It is logically separable and materially more demanding.

No new Lean code or proof objects are supplied, and no Lean build was run. The Python interpreter, its arithmetic implementation, and the supplied program have not been formally verified as part of this package. The mathematical proofs and the finite checks are complementary evidence, not interchangeable forms of certification.

12 Further research questions

The following questions are not conclusions of this paper. They separate natural next steps from claims already established above.

Question 12.1 (Fractional interval and trace-critical spaces). Determine the exact spectrum on $H^s(0, 1)$ and on logarithmically refined interval spaces when s is nonintegral. What occurs at half-integer trace thresholds, and what are the sharp two-term moment estimates for amplitudes whose first unmatched endpoint jet is prescribed? The finite integer-order jet quotient suggests the correct decomposition but does not settle the fractional norm comparison.

Question 12.2 (Dual distributions for higher even moments). For general b, m , each nonzero eigenvector of $A_{b,m}$ should lead to coefficient functionals governed by finite recurrences along

residue classes modulo b . Develop an effective construction, and determine their exact Sobolev and logarithmic thresholds. In the presence of Jordan blocks, identify the additional powers of n in moment expansions and the endpoint logarithmic exponents they force.

Question 12.3 (Which masks admit a positive square-mask eigenfunction?). Characterize the nonzero trigonometric masks satisfying (P). For even sine powers, one-point vanishing and a finite zero count suffice. For masks whose zeros contain periodic inverse-branch configurations, strict positivity may fail. Determine the resulting essential spectral set and whether a decomposition into invariant components replaces the single coisometry similarity.

Question 12.4 (Beyond polynomial masks). For analytic or sufficiently smooth nonpolynomial masks, Fourier conjugacy still suggests a compact leading comparison when the coefficient tails are controlled. Can one recover an exact Sobolev essential disk and classify all exterior eigenvalues through a convergent infinite Fourier reduction? The finite nilpotent quotient used here no longer exists, so its role must be replaced rather than assumed.

Question 12.5 (Nonlinear expanding maps). Replace $T_b x = bx$ by a nonlinear expanding covering of the circle. The Sobolev multiplier ratio then depends on position through the derivative. Find sharp hypotheses under which pressure or a positive square-mask transfer equation determines the entire essential spectral set, not merely an upper bound on its radius. The present proof uses constant expansion in an essential way.

Question 12.6 (Which scalar observables improve at an embedded boundary?). Classify continuous functionals ℓ for which $\ell(\mathcal{L}^n - 2^{-n}P_0 - (-1/4)^n P_1) = o(4^{-n})$ at critical regularity. Integration has an exact low-frequency cutoff. Is a quantitative variant of this cutoff necessary, or can cancellation in other observables give a genuinely different mechanism?

Question 12.7 (Exact constants and dyadic oscillations in the sharp norms). The present estimates determine the exact exponential rate and logarithmic power up to positive constants. Determine whether the corresponding normalized dual norms converge, and when log-periodic oscillations occur for non-dyadic cutoffs. Such a refinement requires weighted distribution information within each annulus, not only its total energy.

Question 12.8 (Quantitative spectral projections near the thresholds). As $s \downarrow s_0$ or $s \downarrow s_1$, the relevant dual norm diverges. The block tests predict the order of this divergence. Determine exact leading constants, bounds uniform in a joint limit of s, β, n , and certified resolvent estimates suitable for stable numerical computation. This would turn the sharp qualitative transition into an effective conditioning theory.

Question 12.9 (Finite-smoothness spaces outside the Hilbert scale). Answer the original repository question on Hölder and C^r spaces. The Hilbert-space coisometry and exact weighted ℓ^2 dual norms are specific to the present proof. Their replacements must account for the relevant Banach norms and cannot be obtained by merely replacing s with an integer differentiability index in the final formula.

Question 12.10 (A proof-producing computational interface). Construct a verified procedure that takes an integer dilation and a rational trigonometric mask, produces the finite matrices and their invariant bands, and checks a supplied positive square-mask eigenfunction certificate. Combining this with a formal proof of [theorem 2.1](#) would turn finite algebraic data into machine-checked infinite-dimensional spectral conclusions for a broad family of explicit examples.

13 Conclusion

The analytic finite spectrum is only one regularity regime of the Rvachev–Thue–Morse transfer operator. On the periodic Sobolev scale, its replacement is exact and computable: a regularity-dependent essential disk together with a finite set of algebraic outliers. The squared sine mask

determines the disk radius, while the unsquared mask determines the finite outliers. These are different pieces of spectral data and require different arguments.

In the dyadic quadratic case, the leading Thue–Morse measure and the subleading Stern distribution have sharp thresholds s_0 and $s_1 = s_0 + 1$. Their exact energy growth explains both spectral isolation and the continuity of asymptotic coefficient functionals. The Fourier-cutoff identity goes further: it identifies the moment remainder as a literal high-frequency tail. This permits a uniform scalar asymptotic at a critical essential boundary, provided and only provided that the subleading functional is made continuous by the logarithmic exponent $\beta > 1/2$. The simultaneous operator-norm obstruction shows why that conclusion could not have been inferred from a spectral gap.

A Repository provenance and claim boundaries

The directly inspected repository materials were the Fabius-function README, the Fourier-decay group README, the spectral-collapse manuscript README and inventory, and the spectral manuscript source at the immutable commit below. The source search located the explicit question *Spectra at finite smoothness*. Inspection was targeted to the relevant research branch, not an exhaustive audit of the whole repository.

Item	Identifier or scope
Repository	VladimirReshetnikov/ProveIt
Immutable comparison commit	b2aed2d23125006c48d63fc97aef0d64b67becb1
Spectral source blob in inspected inventory	1289ae8ca02383bd2a44ad8f9edb429941d004ff
Spectral README blob	7d32919e68332fd8ca78211811b2f005ab97d175
Fourier group README blob	30d7898b50f2bf0ea199cf24f5b543d8b98fe4fe
Source question	Spectra at finite smoothness
Answered here	Periodic $H^{s,\beta}$, all real s, β ; interval H^r , integer $r \geq 0$
Not answered here	Hölder/ C^r spectrum; general fractional interval spaces; full interior point-spectrum decomposition

The spectral source path, relative to the repository root, is

```
Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/
rvachev_up_fourier_decay/Spectral_Collapse_Alternating_RMS/article.tex
```

The observed commit timestamp is 30 September 2026 at 03:29:35 UTC, which is 29 September 2026 at 20:29:35 Pacific daylight time. This accounts for the different UTC and local calendar dates without treating either as a future source.

The earlier manuscript’s analytic determinant and RMS constant enclosures are antecedents. They were not independently recertified here. The present proofs do not rely on their correctness for the new Sobolev or endpoint conclusions. Global priority and peer-review acceptance are not established. The supplied text is intended to be mathematically checkable and is accompanied by an explicit statement of the analytic inputs and finite verification scope.

B Elementary implementation details

The exact field operations for $a + b\sqrt{17}$ use

$$(a, b) + (c, d) = (a + c, b + d), \quad (a, b)(c, d) = (ac + 17bd, ad + bc).$$

All four coordinates are rational numbers. Thus checking (8.4) and (8.5) requires no approximate radical. The two characteristic polynomials are checked by the Faddeev–LeVerrier recurrence over $\mathbb{Q}$, followed by a zero-matrix Cayley–Hamilton check. The finite Riesz products are constructed by multiplying their Laurent coefficient dictionaries, independently of the stable-cutoff formula.

The verification log and JSON output are part of the delivered archive. The defaults are intentionally finite and reproducible. Increasing the level parameter increases the amount of exact arithmetic but does not alter the logical role of computation: the general recurrences and functional-analytic theorems are established by proofs in the article.

References

- [1] Vladimir Reshetnikov, *ProveIt repository, Spectral Collapse and Alternating RMS Asymptotics for the Rvachev Up-Function*, research manuscript dated 28 September 2026, with README. Inspected at commit `b2aed2d23125006c48d63fc97aef0d64b67becb1`. Immutable repository directory. The source question and blob identifiers are recorded in section A.
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- [3] Xianghong Chen and Hans Volkmer, *On transfer operators on the circle with trigonometric weights*, 2016. [arXiv:1610.07658](#).
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- [7] Juan Arias de Reyna, *An infinitely differentiable function with compact support: Definition and Properties*, 2017 English translation of the 1982 paper in *Rev. Real Acad. Ciencias Madrid* **76**, 21–38. [arXiv:1702.05442](#).